

HKCEE 1991 Mathematics II

Label 1 $(a^{2a})(3a^{4a})$

- A. $3a^{6a}$
 B. $(3a)^{6a}$
 C. $3a^{8a}$
 D. $4a^{6a}$
 E. $(3^{4a})(a^{6a})$

Label 2 $\frac{1}{1-x^2} - \frac{1}{(1+x)^2} =$

- A. $\frac{2}{(1-x^2)(1+x^2)}$
 B. $\frac{2x^2}{(1-x^2)(1+x^2)}$
 C. $\frac{(1-x^2)(1+x^2)}{2x^2}$
 D. $\frac{(1-x^2)(1+x)^2}{2}$
 E. $\frac{2}{(1-x)(1+x)^2}$

Label 3 Which one of the following is a factor of $x^3 - 4x^2 + x + 6$?

- A. $(x+1)(x-2)$
 B. $(x+1)(x+2)$
 C. $(x-1)(x+2)$
 D. $(x-1)(x-3)$
 E. $(x-1)(x+3)$

Label 4 If $y = \sqrt{\frac{1+mx}{1-mx}}$, then $x =$

- A. $\frac{m(y-1)}{y+1}$
 B. $\frac{y+1}{y-1}$
 C. $\frac{m(y+1)}{y-1}$

- C. $\frac{(1-y^2)}{m(1+y^2)}$
 D. $\frac{m(y^2-1)}{(y^2+1)}$
 E. $\frac{(y^2-1)}{m(y^2+1)}$

Label 5 $\frac{x^2 + y^2}{\frac{1}{x} + \frac{1}{y}} =$

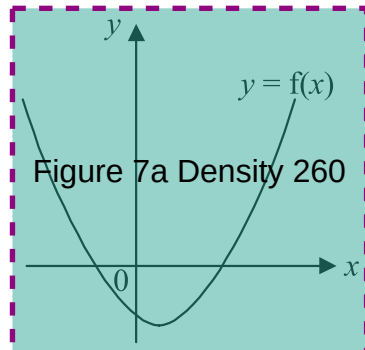
- A. $\frac{1}{x^2} + \frac{1}{y^2}$
 B. $\frac{1}{x^2} + \frac{1}{xy} + \frac{1}{y^2}$
 C. $\frac{1}{x^2} + \frac{2}{xy} + \frac{1}{y^2}$
 D. $\frac{1}{x^2} - \frac{2}{xy} + \frac{1}{y^2}$
 E. $\frac{1}{x^2} - \frac{1}{xy} + \frac{1}{y^2}$

Label 6 The L.C.M. of $x, 2x^2, 3x^3, 4x^4, 5x^5$ is

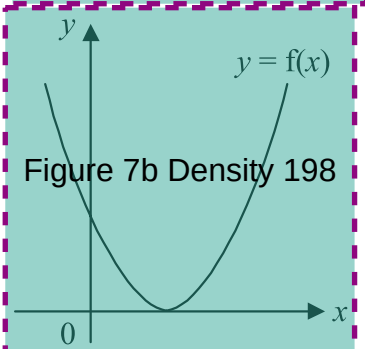
- A. x
 B. $5x^5$
 C. $60x^5$
 D. $120x^5$
 E. $120x^{15}$

Label 7 In which of the following cases the equation $f(x) = 0$ **cannot** be solved by the method of bisection?

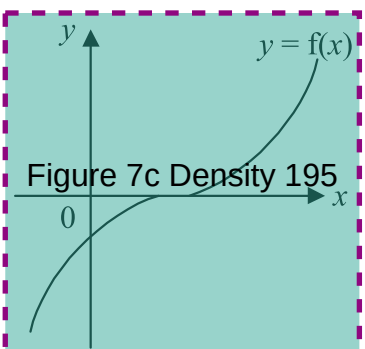
A



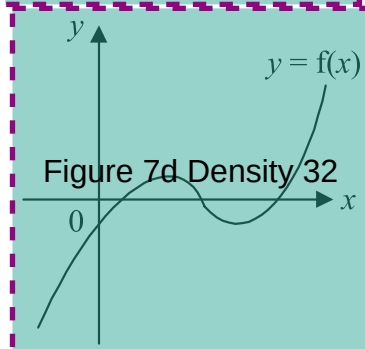
B



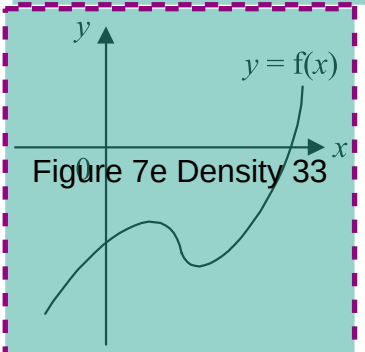
C



D



E



Label 8 Solve the following equations

$$x - 1 = y + 2 = x + y - 3$$

- A. $x = 1, y = -2$
- B. $x = 1, y = 4$
- C. $x = 4, y = 1$
- D. $x = 7, y = -2$
- E. $x = 7, y = 4$

Label 9 Let y vary partly as $\frac{1}{x}$ and partly as x

When $x = 1, y = 5$ and when $x = 4,$
 $y = \frac{25}{2}$. Find y when $x = 2$

- A. $\frac{5}{2}$
- B. 4
- C. $\frac{25}{4}$
- D. 7
- E. $\frac{17}{2}$

Label 10 If $\frac{1}{a} = \frac{1}{b} = \frac{1}{c} = 2$ and $a : c = 4 : 1$, then

$$a : b : c =$$

- A. $12 : 8 : 3$
- B. $8 : 3 : 2$
- C. $4 : 6 : 1$
- D. $2 : 3 : 8$
- E. $2 : 3 : 4$

Label 11 A blanket loses 10% of its length and 8% of its width after washing. The percentage loss in area is

- A. 18.8%
- B. 18%
- C. 17.2%
- D. 9%
- E. 8%

Label 12
12

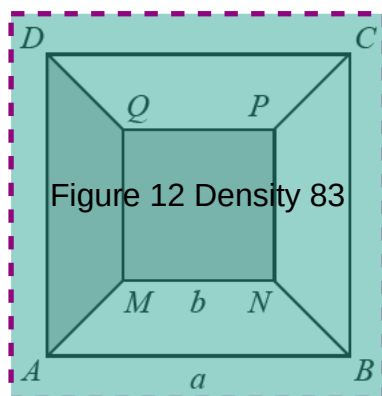


Figure 12 Density 83

In the figure, $ABCD$ is a square of side a and $MNPQ$ is a square of side b . The four trapeziums are identical. The area of the shaded region is

- A. $\frac{3b^2 + a^2}{4}$
- B. $\frac{3b^2 - a^2}{4}$
- C. $\frac{5b^2 + a^2}{4}$
- D. $\frac{5b^2 - a^2}{4}$
- E. $\frac{(a-b)^2}{4} + b^2$

Label 13
13

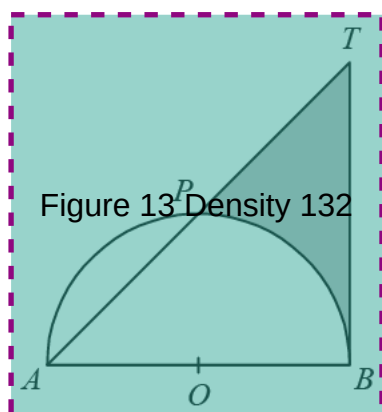


Figure 13 Density 132

In the figure, TB touches the semi-circle at B . TA cuts the semi-circle at P such that $TP = PA$. If the radius of the semi-circle is 2, find the area of the shaded region.

- A. $12 - \pi$
- B. $8 - \pi$

- C. $6 - \pi$
- D. $4 - \pi$
- E. $2(4 - \pi)$

Label 14
14

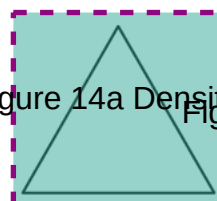


Figure 14a Density 0

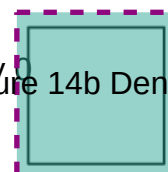


Figure 14b Density 0

An equilateral triangle and a square have equal perimeters.

$$\frac{\text{Area of the triangle}}{\text{Area of the square}} =$$

- A. $\frac{9\sqrt{3}}{16}$
- B. $\frac{\sqrt{3}}{4}$
- C. $\frac{\sqrt{3}}{3}$
- D. $\frac{4\sqrt{3}}{9}$
- E. $\frac{3}{4}$

Label 15
15

A man borrows \$10 000 from a bank at 12% per annum compounded monthly. He repays the bank \$2000 at the end of each month. How much does he still owe the bank just after the second repayment?

- A. \$6181
- B. \$6200
- C. \$6201
- D. \$8304
- E. \$8400

Label 16
16

$$\left[\frac{1}{\cos \theta} + \tan \theta \right] (1 - \sin \theta) =$$

- A. $\sin \theta$
- B. $\cos \theta$
- C. $\cos^2 \theta$
- D. $1 + \sin \theta$
- E. $\sin \theta \tan \theta$

Label 17
17

$$\frac{17 \sin(\theta - 90^\circ)}{\tan(\theta + 180^\circ)} =$$

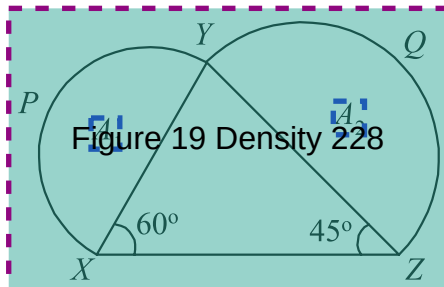
- A. $\cos \theta$
 B. $-\cos \theta$
 C. $\frac{\cos^2 \theta}{\sin \theta}$
 D. $\frac{\cos^2 \theta}{\sin \theta}$
 E. $\frac{1}{\sin \theta}$

Label 18
18

For $0 \leq \theta < 2\pi$, how many roots does the equation $\tan \theta + 2 \sin \theta = 0$ have?

- A. 1
 B. 2
 C. 3
 D. 4
 E. 5

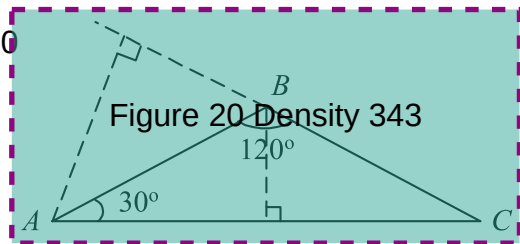
Label 19
19



In the figure, XPY and YQZ are semi-circles with areas A_1 and A_2 respectively. $\angle YXZ = 60^\circ$ and $\angle YZC = 45^\circ$. The ratio $A_1 : A_2 =$

- A. $\sqrt{2} : \sqrt{3}$
 B. $\sqrt{2} : 3$
 C. $2 : 3$
 D. $2 : \sqrt{3}$
 E. $\sqrt{3} : \sqrt{2}$

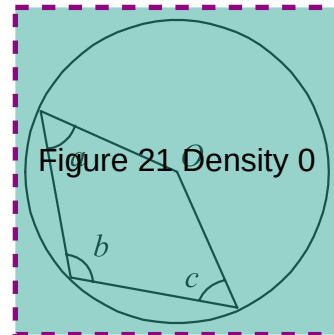
Label 20
20



In the figure, $\angle A = 30^\circ$ and $\angle B = 120^\circ$. The ratio of the altitudes of the triangle ABC from A and from B is

- A. $2 : 1$
 B. $\sqrt{3} : 1$
 C. $\sqrt{2} : 1$
 D. $1 : \sqrt{2}$
 E. $1 : \sqrt{3}$

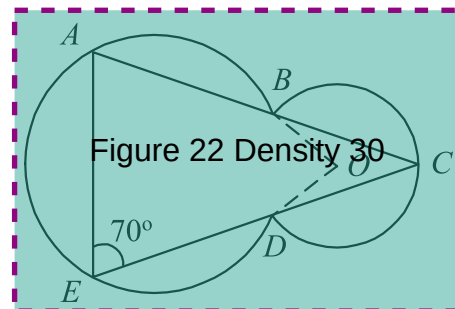
Label 21
21



In the figure, O is the centre of the circle. Find $a + c$.

- A. b
 B. $2b$
 C. $180^\circ - b$
 D. $360^\circ - b$
 E. $360^\circ - 2b$

Label 22
22

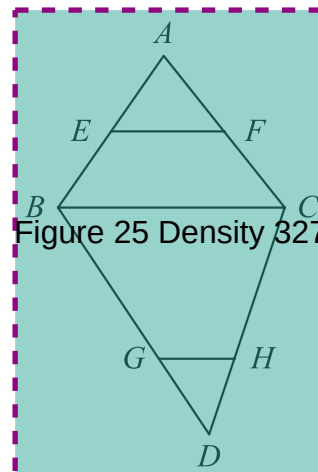


In the figure, O is the centre of the circle BCD . ABC and EDC are straight lines. $BC = DC$ and $\angle AED = 70^\circ$. Find $\angle BOD$.

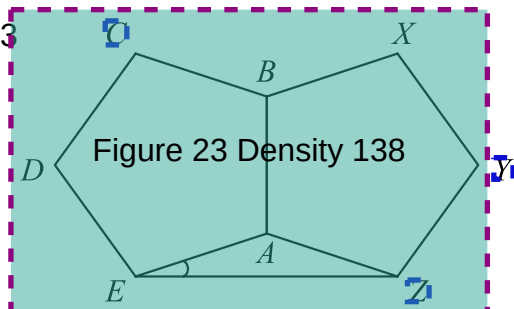
- A. 40°
- B. 70°
- C. 80°
- D. 90°
- E. 140°

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

Label 25



Label 23



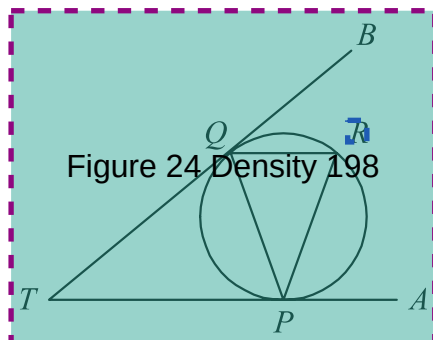
In the figure, $ABCDE$ and $ABXYZ$ are two identical regular pentagons. Find $\angle AEZ$.

- A. 15°
- B. 18°
- C. 24°
- D. 30°
- E. 36°

In the figure, E and F are the mid-points of AB and AC respectively. G and H divide DB and DC respectively in the ratio $1 : 3$. If $EF = 12$, find GH .

- A. 3
- B. 4
- C. 6
- D. 8
- E. 12

Label 24



In the figure, TPA and TQB are tangents to the circle at P and Q respectively. If $PQ = PR$, which of the following **must** be true?

- I. $\angle APR = \angle QRP$
- II. $\angle QTP = \angle QPR$
- III. $\angle QPR = \angle APR$

Label 26

The circle $x^2 + y^2 + 4x + ky + 4 = 0$ passes through the point $(1, 3)$. The radius of the circle is

- A. $\sqrt{68}$
- B. $\sqrt{48}$
- C. $\sqrt{17}$
- D. 6
- E. 3

Label 27

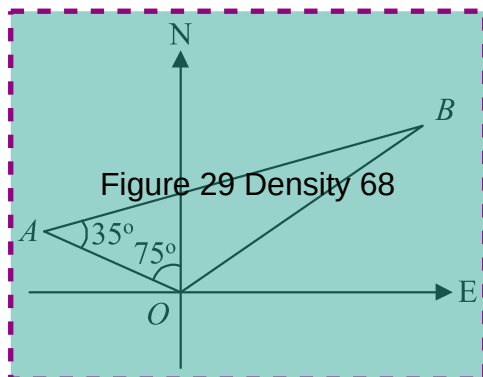
Let A and B be the points $(4, -7)$ and $(-6, 5)$ respectively. The equation of the line passing through the mid-point of AB and perpendicular to $3x - 4y + 14 = 0$ is

- A. $3x - 4y - 1 = 0$
 B. $3x + 4y + 7 = 0$
 C. $4x - 3y + 1 = 0$
 D. $4x + 3y - 7 = 0$
 E. $4x + 3y + 7 = 0$

Label 28
28
PQRS is a parallelogram with vertices $P = (0, 0)$, $Q = (a, b)$ and $S = (-b, a)$. Find R .

- A. $(-a, -b)$
 B. $(a, -b)$
 C. $(a - b, a - b)$
 D. $(a - b, a + b)$
 E. $(a + b, a + b)$

Label 29
29



In the figure, A and B are the positions of two boats. The bearing of B from A is

- A. $N55^\circ E$
 B. $N70^\circ E$
 C. $N20^\circ E$
 D. $S35^\circ E$
 E. $S75^\circ E$

Label 30
30
The mean and standard deviation of a distribution of test scores are m and s respectively. If 4 marks are added to each score of the distribution, what are the mean and standard deviation of the new distribution?

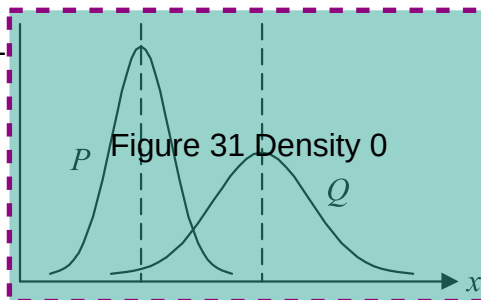
Mean

Standard
Deviation

- A. $m + 4$ s
 B. $m + 4$ $s + 4$
 C. $m + 4$ $s + 4$

- D. m $s + 2$
 E. m $s + 4$

Label 31
31



The graph shows the frequency curves of two symmetric distributions P and Q . Which of the following is /are true?

- I. The mean of $P <$ the mean of Q .
 II. The mode of $P >$ the mode of Q .
 III. The inter-quartile range of $P <$ the inter-quartile range of Q .

- A. I only
 B. I and II only
 C. I and III only
 D. II and III only
 E. I, II and III

Label 32
32
A fair die is thrown 3 times. The probability that "6" occurs exactly once is

- A. $\frac{1}{3}$
 B. $\left(\frac{1}{6}\right)^3$
 C. $\frac{1}{3} \times \frac{1}{6}$
 D. $\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2$
 E. $3\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2$

Label 33
33
If $(\sqrt{3} + 1)\sqrt{x} = 2$, then $x =$

- A. $2 - \sqrt{3}$

- B. $\sqrt{3}-1$
 C. $\frac{1}{2}$
 D. $2(2-\sqrt{3})$
 E. $4-\sqrt{3}$

Label 34 If $\log x : \log y = m : n$, then $x =$

- A. $\frac{my}{n}$
 B. $(m-n)y$
 C. $m-n+y$
 D. $\frac{m}{y^n}$
 E. $\frac{m \log y}{n}$

Label 35 If $f(x) = x - \frac{1}{x}$, then $f(x) - f\left(\frac{1}{x}\right) =$

- A. 0
 B. $2x$
 C. $-\frac{2}{x}$
 D. $2\left(x - \frac{1}{x}\right)$
 E. $2\left(\frac{1}{x} - x\right)$

Label 36 If $p(x^2 - x) + q(x^2 + x) = 4x^2 + 8x$, find p and q .

- A. $p = 4, q = 8$
 B. $p = -8, q = 4$
 C. $p = -2, q = 6$
 D. $p = 2, q = 6$
 E. $p = 6, q = -2$

Label 37 If $x < 0 < y$, then which one of the following **must** be positive?

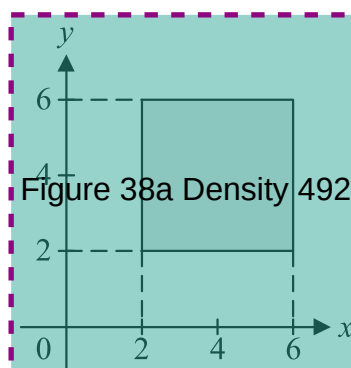
- A. $x+y$
 B. $x-y$
 C. $y-x$
 D. xy

- E. $\frac{y}{x}$

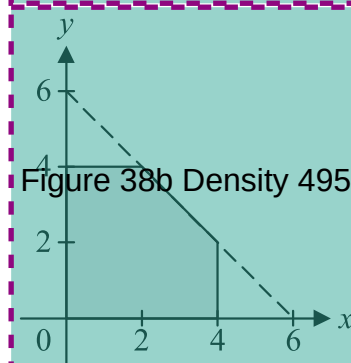
Label 38 Which one of the following shaded regions represents the solution of

$$\begin{cases} 2 \leq x+y \leq 6 \\ 0 \leq x \leq 4 \\ 0 \leq y \leq 4 \end{cases}$$

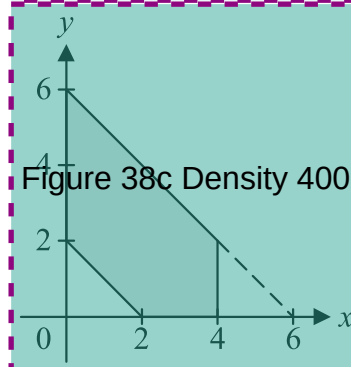
A.



B.



C.



D.

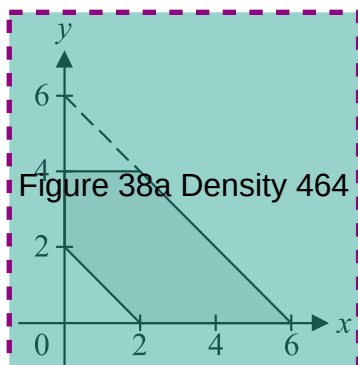


Figure 38a Density 464

E.

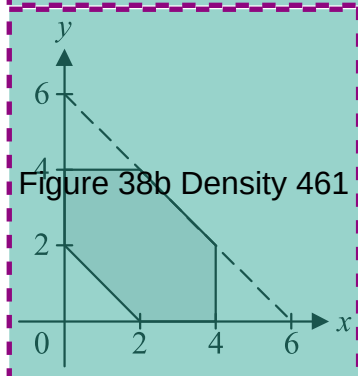


Figure 38b Density 461

Label 39 If $(x - 2)(x - 3) = (a - 2)(a - 3)$, solve for x .

- A. $x = 0$ or 5
- B. $x = 2$ or 3
- C. $x = a$ or 2
- D. $x = a$ or 3
- E. $x = a$ or $5 - a$

Label 40 If the sum to n terms of an A.P. is $n^2 + 3n$, find the 7th term of the A.P.

- A. 16
- B. 18
- C. 54
- D. 70
- E. It cannot be found.

Label 41 If x, y, z are in G.P., which of the following **must** be true?

- I. $x + 3, y + 3, z + 3$ are in G.P.
- II. $3x, 3y, 3z$ are in G.P.
- III. x^2, y^2, z^2 are in G.P.

- A. I only
- B. II only
- C. III only

- D. I and II only
- E. II and III only

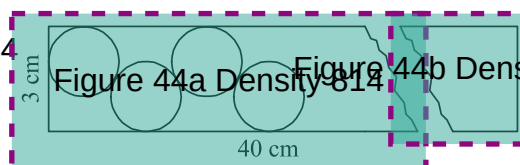
Label 42 2 kg of a solution contains 40% of alcohol by weight. How much alcohol should be added to obtain a solution containing 50% of alcohol by weight?

- A. 0.3 kg
- B. 0.6 kg
- C. 0.75 kg
- D. 1.5 kg
- E. 3.75 kg

Label 43 P sold an article to Q at a profit of 25%. Q sold it to R also at a profit of 25%. If Q gained \$500, how much did P gain?

- A. \$250
- B. \$320
- C. \$333
- D. \$400
- E. \$500

Label 44

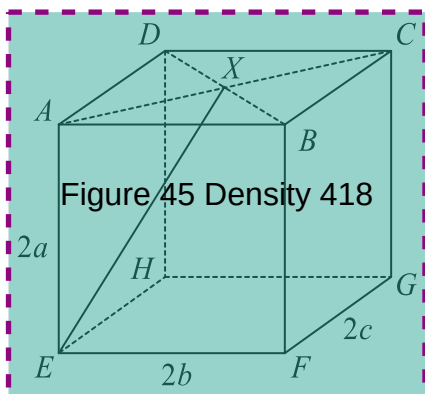


From a rectangular metal sheet of width 3 cm and length 40 cm, at most how many circles each of radius 1 cm can be cut?

- A. 20
- B. 21
- C. 22
- D. 23
- E. 24

DIRECTIONS: Question 45 and 46 refer to the figure below, which shows a cuboid $ABCDEFGH$ with $AE = 2a$, $EF = 2b$ and $FG = 2c$. AC and BD intersect at X .

Label 45
45



Label 46
46

- A. $\sqrt{a^2 + b^2 + c^2}$
 B. $\sqrt{a^2 + b^2 + (2c)^2}$
 C. $\sqrt{a^2 + (2b)^2 + c^2}$
 D. $\sqrt{(2a)^2 + b^2 + c^2}$
 E. $2\sqrt{a^2 + b^2 + c^2}$

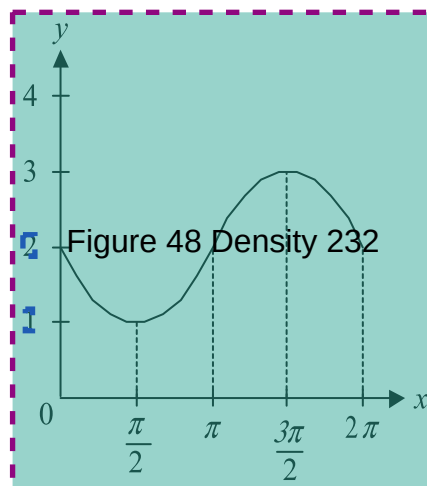
Label 46
46 If the angle between XE and the plane $EFGH$ is θ , then $\tan \theta =$

- A. $\frac{a}{b}$
 B. $\frac{2a}{b}$
 C. $\frac{\sqrt{(2a)^2 + c^2}}{b}$
 D. $\frac{a}{\sqrt{b^2 + c^2}}$
 E. $\frac{2a}{\sqrt{b^2 + c^2}}$

Label 47
47 $\cos \frac{\pi}{2} + \cos \pi + \cos \frac{3\pi}{2} + \cos 2\pi + \dots + \cos 10\pi =$

- A. 0
 B. 1
 C. -1
 D. 10
 E. -10

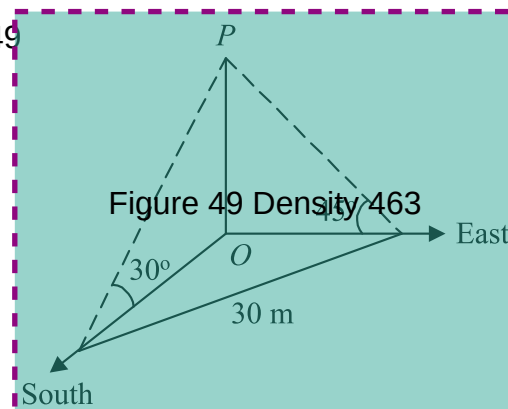
Label 48
48



The figure shows the graph of the function

- A. $y = 2 \cos x$
 B. $y = 2 - \sin x$
 C. $y = 2 + \sin x$
 D. $y = 2 - \cos x$
 E. $y = 2 + \cos x$

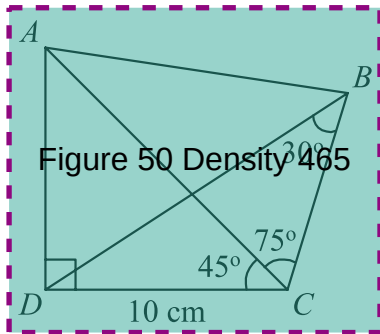
Label 49
49



In the figure, the height of the vertical pole PO is

- A. 7.5 m.
 B. 15 m.
 C. $15\sqrt{2}$ m.
 D. $15\sqrt{3}$ m.
 E. 45 m.

Label 50
50



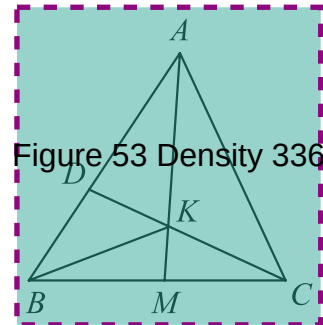
In the figure, arc AB : arc BC : arc CD : arc DE : arc $EA = 1 : 2 : 3 : 4 : 5$. Find θ .

- A. 30°
- B. 36°
- C. 60°
- D. 72°
- E. 120°

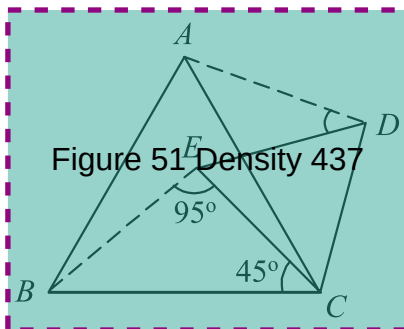
In the figure, find the length of AB , correct to the nearest cm.

Label 53
53

- A. 14 cm
- B. 15 cm
- C. 16 cm
- D. 17 cm
- E. 18 cm



Label 51
51



In the figure, M is the mid-point of BC and $AD = 2DB$. AM and CD intersect at K . Find $\frac{\text{area of } \triangle ADK}{\text{area of } \triangle AKC}$.

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

In the figure, ABC and CDE are equilateral triangles. Find $\angle ADE$.

- A. 15°
- B. 35°
- C. 40°
- D. 45°
- E. 50°

Label 54
54

In the figure, which of the pairs of triangles **must** be congruent?

Label 52
52

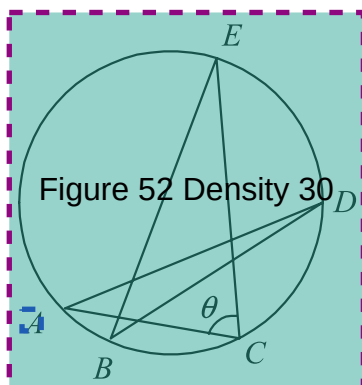
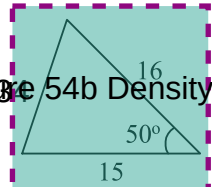
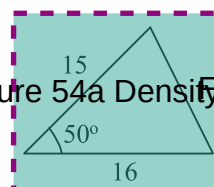
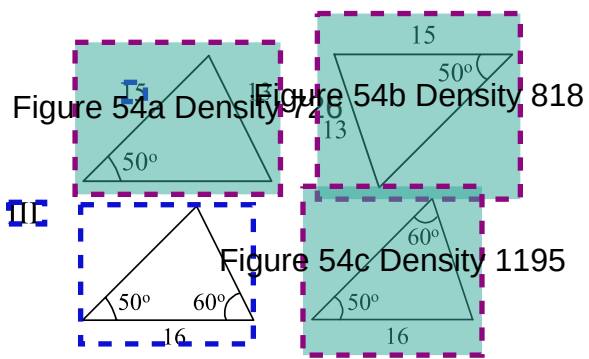


Figure 54a



Q1



Q2

- A. I only
- B. II only
- C. I and III only
- D. II and III only
- E. I, II and III